

Aryan Sadvelkar

919 Westcott Street, Syracuse, New York 13210

asadvelk@syr.edu | 680-207-9488 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Professional Summary

Software engineer building scalable backend systems, developer tools, and AI-powered applications with Python, Node.js, FastAPI, PostgreSQL, Docker, and CI/CD.

Technical Skills

Languages:	Python, JavaScript, TypeScript, SQL, Rust
AI / ML:	PyTorch, TensorFlow, Scikit-learn, LangGraph, LLM APIs (Groq/OpenAI), NLP, OCR
Agentic AI:	Multi-Agent Systems, ReAct Pattern, LangGraph StateGraph, AutoML, Self-Healing Pipelines
Backend:	Node.js, Express.js, Flask, FastAPI, RESTful APIs, Microservices
Frontend:	React.js, HTML5, CSS3, Responsive Design
Databases:	PostgreSQL, MySQL, MongoDB, Query Optimization, Indexing
Cloud / DevOps:	AWS (EC2, S3, Lambda), Docker, CI/CD, Git, GitHub Actions

Education

Master of Science in Computer Science, Syracuse University *Aug 2024 – May 2026*
Coursework: Algorithms, Operating Systems, Computer Architecture, Artificial Intelligence, Agentic AI *CGPA: 3.75*

Bachelor of Engineering in Computer Science, University of Mumbai *Aug 2020 – May 2024*
Honors in AI & ML | Coursework: Software Engineering, Database Management, Data Science *CGPA: 3.62*

Professional Experience

Software Developer Intern, iConsult Collaborative – Syracuse University *Oct 2025 – Present*

- Developed and maintained 15+ RESTful API endpoints for user, admin, and payment workflows using Node.js, Express.js, and PostgreSQL, following microservices design principles
- Reduced average API response time by 25% through SQL query optimization, strategic indexing, and connection pooling
- Implemented JWT-based authentication and role-based access control (RBAC) for secure, scalable user authorization
- Integrated Stripe payment APIs and webhook event handling to support fault-tolerant transaction processing
- Containerized services using Docker and configured CI/CD deployment pipelines for consistent dev and production environments

Projects

VerifAI-ML – Verifiable Multi-Agent ML Engineering System

- Designed and implemented a six-agent ReAct system using LangGraph StateGraph to autonomously build, verify, and self-heal machine learning pipelines for tabular data, with no human intervention required
- Integrated Llama 3.3 70B via the Groq API for dynamic model selection across 12 candidate algorithms, LLM-driven feature engineering, preprocessing strategy decisions, and natural language pipeline explanations
- Built a self-healing feedback loop that detects data leakage and overfitting, generates structured fix plans, and automatically retrains via LangGraph conditional edge routing for up to three retry cycles
- Implemented a shared blackboard memory architecture (TypedDict) enabling fully auditable, modular multi-agent coordination; achieved 0.8156 accuracy, 0.8142 F1, and 0.8434 ROC-AUC on the Titanic benchmark
- Engineered five defense-in-depth security guardrails (input validation, leakage detection, anomaly detection, policy enforcement, execution monitoring) and auto-generated a model card after every pipeline run

CodePulse – AI-Powered GitHub Repository Analyzer

[Live Link](#)

- Built a full-stack web application using React (frontend) and Python/FastAPI (backend) to analyze GitHub repositories for code quality, activity trends, and maintainability scores
- Designed interactive dashboards featuring health scores, commit heatmaps, and contributor analytics, improving developer visibility into repository health
- Implemented backend analysis workflows to detect complexity spikes, oversized files, and risk indicators across large codebases; delivered actionable insights through clear data visualizations

NutriLens AI – AI/OCR Food Label Extraction

- Engineered an end-to-end AI pipeline to extract and structure nutritional data from food label images using OCR and large language model-based parsing; processed and validated 100+ samples into structured JSON
- Reduced extraction formatting errors by ~20% through iterative prompt engineering and output validation logic; built entity extraction to identify nutrients, quantities, and dietary attributes from unstructured text

Publications

Revolutionizing Student Accommodation: A MERN-Based Rental System for Seamless Property Management
International Journal for Multidisciplinary Research, 2024 | [Publication Link](#)